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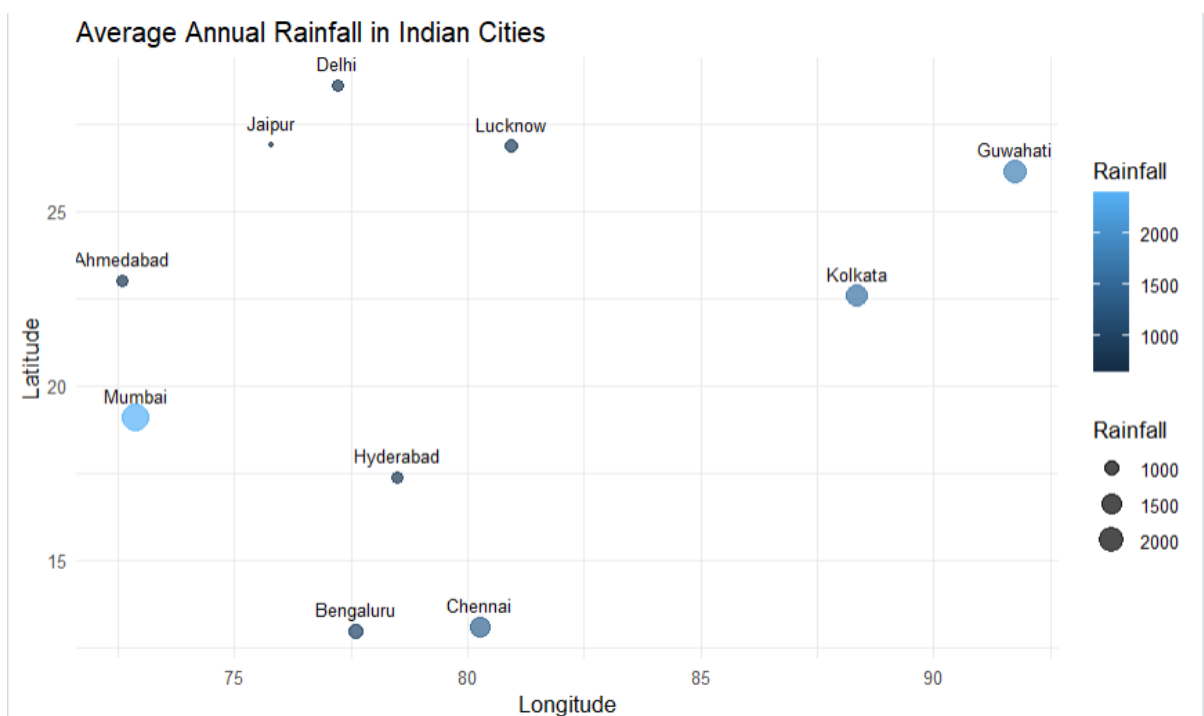
REG NO;192424297

- 1) [Geospatial Data] Given the average annual rainfall (mm) recorded in 10 cities across India along with their locations, construct a suitable map-based visualization.

Code;

```
# Load library
library(ggplot2)
# Data
rainfall <- data.frame(
  City = c("Delhi", "Mumbai", "Chennai", "Kolkata", "Bengaluru",
           "Hyderabad", "Ahmedabad", "Jaipur", "Lucknow", "Guwahati"),
  Longitude = c(77.21, 72.88, 80.27, 88.36, 77.59, 78.49, 72.57, 75.79, 80.95, 91.74),
  Latitude = c(28.61, 19.08, 13.08, 22.57, 12.97, 17.39, 23.02, 26.91, 26.85, 26.14),
  Rainfall = c(800, 2400, 1400, 1600, 970, 790, 780, 650, 900, 1800)
)
# Bubble map
ggplot(rainfall, aes(x = Longitude, y = Latitude)) +
  geom_point(aes(size = Rainfall, color = Rainfall), alpha = 0.7) +
  geom_text(aes(label = City), vjust = -1, size = 3) +
  labs(
    title = "Average Annual Rainfall in Indian Cities",
    x = "Longitude",
    y = "Latitude"
  ) +
  theme_minimal()
```

OUTPUT;



- 2) [Network Visualizations] Given a list of flight routes connecting 9 airports, construct a network diagram showing which airports are directly connected.

Code;

```
# Install package if needed
```

```
# install.packages("igraph")
```

```
library(igraph)
```

```
# Flight routes between 9 airports
```

```
routes <- data.frame(
```

```
  From = c("Delhi", "Delhi", "Mumbai", "Mumbai", "Chennai",  
           "Kolkata", "Hyderabad", "Bengaluru", "Pune", "Delhi"),
```

```
  To = c("Mumbai", "Chennai", "Bengaluru", "Hyderabad", "Kolkata",  
         "Delhi", "Pune", "Chennai", "Delhi", "Hyderabad")
```

```
)
```

```
# Create graph
```

```
g <- graph_from_data_frame(routes, directed = FALSE)
```

```
# Plot network diagram
```

```
plot(
```

```
  g,
```

```
  vertex.size = 30,
```

```
  vertex.color = "skyblue",
```

```
  vertex.label.color = "black",
```

```
  vertex.label.cex = 0.9,
```

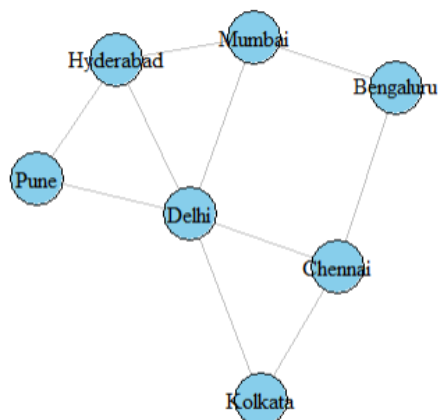
```
  edge.color = "gray",
```

```
  main = "Flight Routes Connecting 9 Airports"
```

```
)
```

Output;

Flight Routes Connecting 9 Airports



- 3) Temporal Data Representations] Given the quarterly profit (₹ crores) of a company over the last 5 years, construct an appropriate time- series chart.

Code;

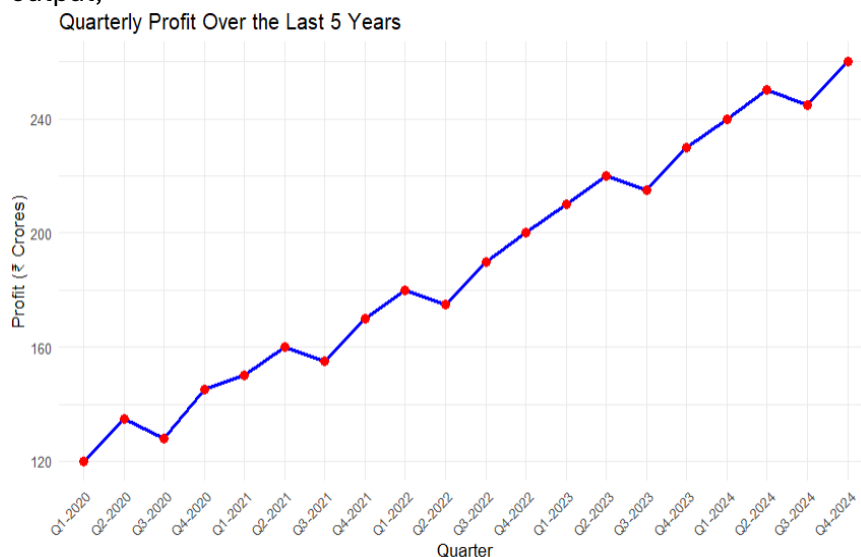
```
# Load library
library(ggplot2)

# Create data
profit <- data.frame(
  Quarter = c("Q1-2020", "Q2-2020", "Q3-2020", "Q4-2020",
    "Q1-2021", "Q2-2021", "Q3-2021", "Q4-2021",
    "Q1-2022", "Q2-2022", "Q3-2022", "Q4-2022",
    "Q1-2023", "Q2-2023", "Q3-2023", "Q4-2023",
    "Q1-2024", "Q2-2024", "Q3-2024", "Q4-2024"),
  Profit = c(120, 135, 128, 145,
    150, 160, 155, 170,
    180, 175, 190, 200,
    210, 220, 215, 230,
    240, 250, 245, 260)
)

# Convert Quarter to ordered factor
profit$Quarter <- factor(profit$Quarter, levels = profit$Quarter)

# Create time-series line chart
ggplot(profit, aes(x = Quarter, y = Profit, group = 1)) +
  geom_line(color = "blue", linewidth = 1) +
  geom_point(color = "red", size = 2.5) +
  labs(
    title = "Quarterly Profit Over the Last 5 Years",
    x = "Quarter",
    y = "Profit (₹ Crores)"
  ) +
  theme_minimal() +
  theme(axis.text.x = element_text(angle = 45, hjust = 1))
```

output;



- 4) [Multivariate Data Visualization] Given data on 20 employees with variable department,salary, years of experience, and performance rating, construct an appropriate multivariate visualization.

Code;

```
# Load library
```

```
library(ggplot2)
```

```
# Create data
```

```
employees <- data.frame(
```

```
  Department = c("HR","HR","IT","IT","Sales","Sales","Finance","Finance",  
                 "HR","IT","Sales","Finance","HR","IT","Sales","Finance",  
                 "HR","IT","Sales","Finance"),
```

```
  Salary = c(35000,40000,55000,60000,45000,50000,65000,70000,  
            42000,58000,47000,68000,39000,62000,52000,72000,  
            41000,64000,53000,75000),
```

```
  Experience = c(2,4,5,7,3,6,8,10,3,6,4,9,2,8,5,11,4,9,6,12),
```

```
  Performance = c(3,4,4,5,3,4,5,5,4,5,3,4,3,5,4,5,4,5,4,5)
```

```
)
```

```
# Create bubble chart
```

```
ggplot(employees,
```

```
  aes(x = Experience,
```

```
    y = Salary,
```

```
    color = Department,
```

```
    size = Performance)) +
```

```
geom_point(alpha = 0.7) +
```

```
labs(
```

```
  title = "Employee Salary, Experience and Performance",
```

```
  x = "Years of Experience",
```

```
  y = "Salary (₹)"
```

```
) +
```

```
theme_minimal()
```

```
output;
```



- 5) [Bar Plots] Given the number of employees working in 5 departments of a company, construct a bar plot to visualize the data.

Code;

```
# Load library
```

```
library(ggplot2)
```

```
# Create data
```

```
employees <- data.frame(
```

```
  Department = c("HR", "Finance", "IT", "Sales", "Marketing"),
```

```
  Employees = c(25, 18, 40, 30, 22)
```

```
)
```

```
# Create bar plot
```

```
ggplot(employees, aes(x = Department, y = Employees, fill = Department)) +
```

```
  geom_bar(stat = "identity") +
```

```
  labs(
```

```
    title = "Number of Employees in Each Department",
```

```
    x = "Department",
```

```
    y = "Number of Employees"
```

```
  ) +
```

```
  theme_minimal()
```

```
output;
```

